Design Specification

Submission for

Internet Technology COMPSCI-5012
by Group CK

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Project overview of our Uptime Monitor WebApp:

Upty

We are making an **Uptime Monitor WebApp.**

This will be a web application that monitors the availability of websites and APIs by periodically checking if they respond correctly.

When a service goes down, the system logs the incident and notifies the team.

Main value/ benef for users:

- Professional monitoring without the high cost
- Built specifically for small teams and startups
- Simple enough for non-technical users
- Reliable real-time alerts
- Setup in minutes, not hours

Intended Users:

- Small to medium development teams,
- DevOps engineers, and system administrators
- Startup founders managing their own infrastructure

Problems it addresses:

- Teams often discover their services are down only when customers complain.
- Manual checking is unreliable and time-consuming.
- Existing solutions are expensive and complex for small teams.

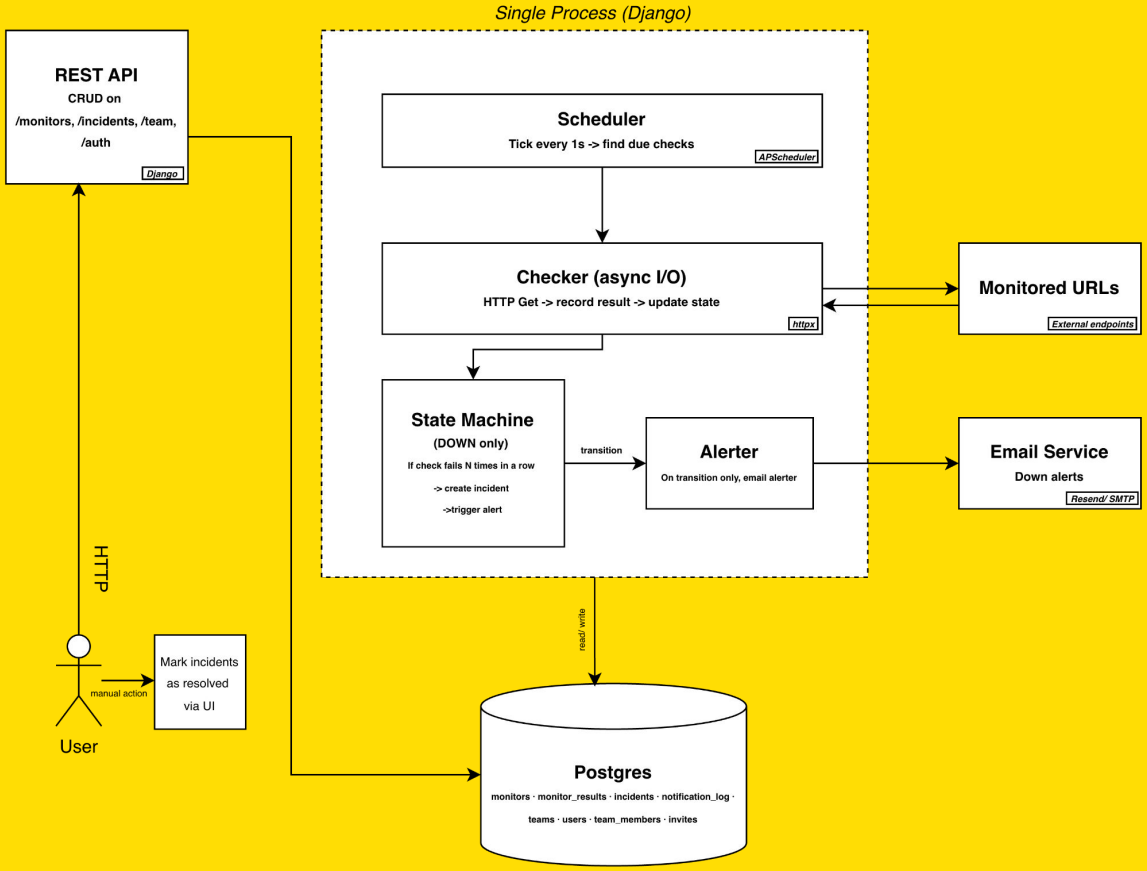
Concept Flow:

Register/login → land on dashboard
Create monitors → they start getting checked
Something goes down → you get an email, incident appears on /incidents
Fix the problem → mark it resolved
Invite teammates → they can see everything too

User Stories

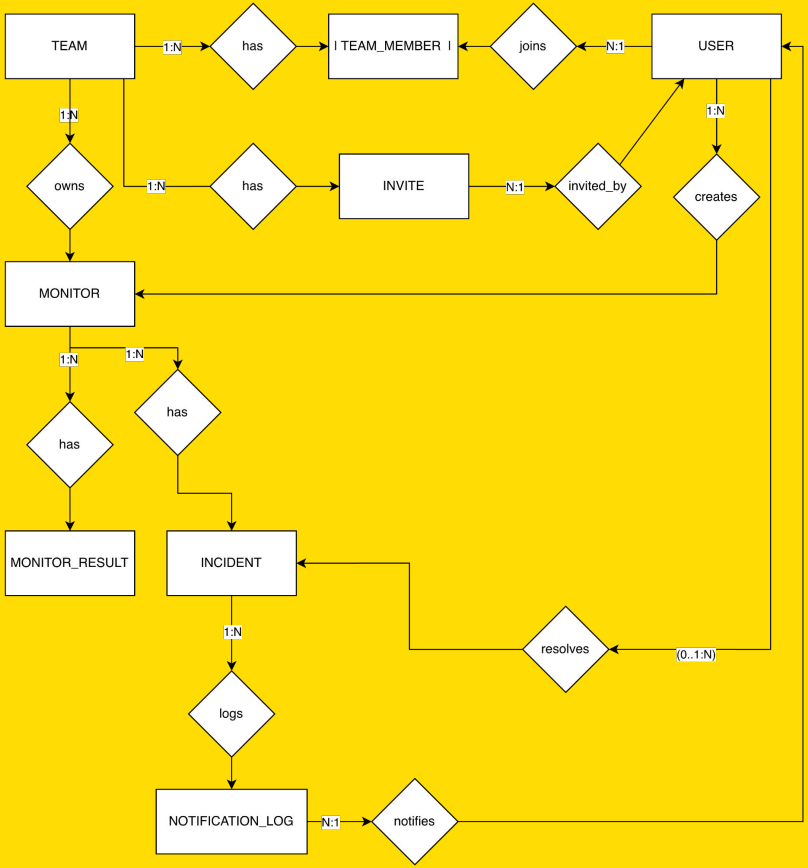
Story No.	Title	User Story	Priority	Cost	Acceptance Tests
1	Automatic Incident Creation	As a user, I want incidents to be automatically created when a monitor goes down, so that I have a record of outages	Must Have	2	❑ Incident is created automatically when monitor transitions from UP to DOWN ❑ Incident record includes monitor ID, start time, and status ❑ No duplicate incidents created for consecutive failed checks
2	Register and Log In	As a user, I want to register and log in, so that I can access my monitors	Must Have	1	❑ User can create an account with email and password, then is redirected to dashboard ❑ User can log in with valid credentials and sees their dashboard ❑ Invalid login shows an error message without revealing which field is wrong
3	Create and Join Teams	As a user, I want to create and join teams, so that I can collaborate with my teammates	Must Have	2	❑ New user registration creates a team automatically ❑ Existing user can join a team via invite link ❑ Team page shows all current team members
4	Add and Delete Monitors	As a user, I want to add a new/delete monitor by entering a URL, so that its uptime starts to be tracked	Must Have	1	❑ User can create a monitor by entering a valid URL ❑ New monitor appears on dashboard after creation ❑ User can delete a monitor and it no longer appears on dashboard
5	Automatic Monitor Checks	As a user, I want monitors to be checked automatically at regular intervals, so that I don't have to manually check each one	Must Have	3	❑ New monitor receives its first check within 60 seconds of creation ❑ Monitor status updates automatically based on check results ❑ Check results are recorded with timestamp and response status
6	View Dashboard	As a team member, I want to view a dashboard showing all monitors and their statuses, so that I can see health at a glance	Must Have	2	❑ Dashboard displays all team monitors with current UP/DOWN status ❑ DOWN monitors are visually distinct from UP monitors ❑ Empty dashboard shows prompt to create first monitor
7	Resolve Incidents	As a user, I want to mark incidents as resolved, so that I can track which issues have been addressed	Should Have	1	❑ User can mark an open incident as resolved ❑ Resolved incidents show resolution timestamp ❑ Resolved incidents are visually distinct from open incidents
8	Add Alert Contacts	As a team member, I want to add alert contacts, so that the right people are notified	Should Have	1	❑ User can add an email address as an alert contact ❑ Alert contact is linked to one or more monitors ❑ User can remove an alert contact
9	View Notification Log	As a team member, I want to view a log of all notifications sent, so that I can see what alerts were triggered	Should Have	1	❑ Notification log shows all sent alerts with timestamp ❑ Each log entry shows which monitor and contact were involved ❑ Log entries are sorted with most recent first
12	Manage Account Settings	As a user, I want to update my email, password, and API key, so that I can keep my account secure and current	Should Have	2	❑ User can update email and password from settings page ❑ User can view (masked) and copy their API key ❑ Regenerating API key invalidates the old one immediately
11	Email Notifications	As a user, I want to receive email notifications when alerts arrive, so that I don't miss critical issues	Could Have	3	❑ Email is sent to alert contact when monitor goes DOWN ❑ Email is sent when monitor recovers (goes UP) ❑ Email contains monitor URL and current status
12	Team Roles	As a team admin, I want to assign roles like admin/member, so that I can control who can modify settings	Could Have	2	❑ Team creator is automatically assigned admin role ❑ Admin can change another member's role ❑ Members cannot access admin-only actions (like deleting team)

System Architecture Diagram (single process, single database)



Link to diagram (made using draw.io): https://drive.google.com/file/d/1D-nQKtFUEB1vfxC_vsfGNsTWzaY-YVj-/view?usp=drive_link

Entity Relationship Diagram



Link to diagram (made using draw.io): https://app.diagrams.net/#G1QWpSEi5XXqGXGNn2XKvvKyxDwYsECnff#%7B%22pageId%22%3A%22pWeCYvARtkCKhdSpNGMT%22%7D

Entity Attributes & Descriptions

Core Entities

Entity	Attributes	Description
TEAM	id, name, created_at	Workspace that groups users and monitors together
USER	id, email, password, created_at	Account info; email doubles as alert destination
TEAM_MEMBER	id, team_id (FK), user_id (FK), joined_at	Resolves many-to-many between Team and User

Monitoring Entities

Entity	Attributes	Description
MONITOR	id, team_id (FK), created_by (FK), url, failure_threshold, current_state, last_checked_at, created_at	URL being monitored with check settings; created_by determines who receives alerts
MONITOR_RESULT	id, monitor_id (FK), status_code, checked_at	Individual check result history

Incident & Alert Entities

Entity	Attributes	Description
INCIDENT	id, monitor_id (FK), started_at, is_resolved, resolved_by (FK),	Outage record; resolved manually by user
NOTIFICATION_LOG	id, incident_id (FK), user_id (FK), sent_at, status	Audit trail of alert emails sent

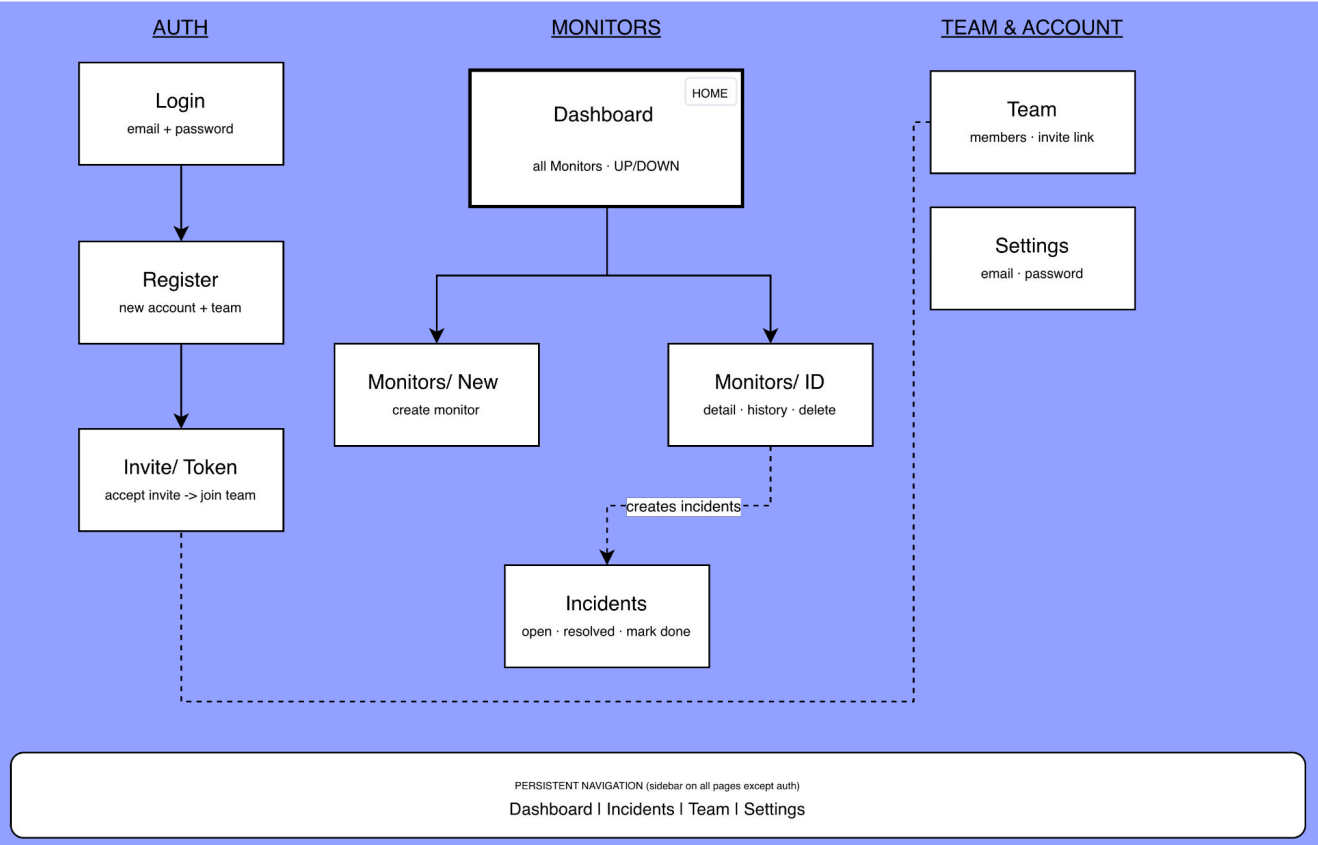
Invite Entity

Entity	Attributes	Description
INVITE	id, team_id (FK), invited_by (FK), token, expires_at, accepted, created_at	Team invitation with expiring token

Key Constraints

- TEAM_MEMBER: Unique constraint on (team_id, user_id) prevents duplicate memberships
- MONITOR.created_by: The user who created the monitor receives all alerts for it
- INCIDENT.resolved_by: only populated when a user manually resolves the incident, null till then
- INVITE.token: Unique constraint for fast lookup

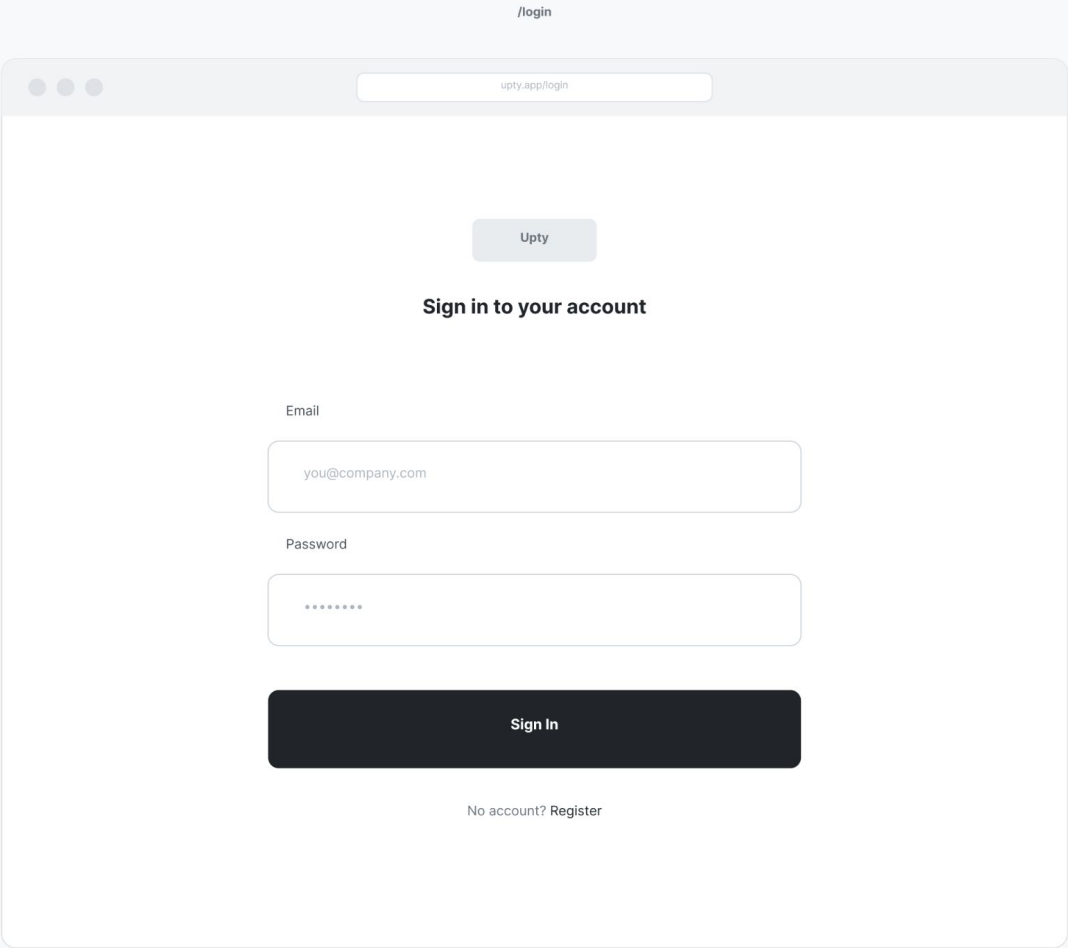
Site Map



9 pages, 3 sections

Link to diagram (made using draw.io): https://app.diagrams.net/#G1UODWS3bSVT5K4SUvk31hzGSEtEuEbCEE#%7B%22pageId%22%3A%22p1omMMYIVWV5MQ757zdf%22%7D

Wireframe 1



Link to diagram (made using figma): https://www.figma.com/site/CFWRT6lpKymj6Xml8Yv2gK/Internet-tech-project?node-id=0-1&t=8FYUEBdoJ2UNnWUf-1

Wireframe 2

/dashboard

upty.app/dashboard

Upty

Dashboard

Incidents

Team

Settings

Monitors

3 monitors · 2 up · 1 down

+ New Monitor

URL	STATUS	LATENCY	LAST CHECK
staging.mysite.com /health	DOWN	—	3m ago
api.example.com /health	UP	124ms	12s ago
www.mysite.com /	UP	89ms	45s ago

Each row is clickable → /monitors/:id

Sorted: DOWN first, then by last checked

EMPTY STATE:

No monitors yet.

+ Add your first monitor

Link to diagram (made using figma): https://www.figma.com/site/CFWRT6lpKymj6Xml8Yv2gK/Internet-tech-project?node-id=0-1&t=8FYUEBdoJ2UNnWUf-1

Wireframe 3

/monitors/:id

upty.app/monitors/abc-123

Upty

Dashboard

Incidents

Team

Settings

← Back to Dashboard

api.example.com/health

UP

Delete

STATUS	LAST CHECK	LATENCY
UP	12s ago	124ms

Settings

Check every: 60s Timeout: 5000ms Expected: 200 Failures before alert: 3

Alerts go to: you@company.com (you)

Recent Results

TIME	STATUS	LATENCY	RESULT
2026-02-01 14:32:01	200	124ms	✓
2026-02-01 14:31:01	200	118ms	✓
2026-02-01 14:30:01	503	4820ms	✕ Connection timeout
2026-02-01 14:29:01	200	95ms	✓

Load more ↓

Link to diagram (made using figma): https://www.figma.com/site/CFWRT6lpKymj6Xml8Yv2gK/Internet-tech-project?node-id=0-1&t=8FYUEBdoJ2UNnWUf-1

Wireframe 4

/incidents

upty.app/incidents

Upty

Dashboard

Incidents

Team

Settings

Incidents

2 open · 3 resolved

Open

staging.mysite.com
Down since Feb 1, 2026 at 14:04
Duration: 28 minutes
Resolve

api.example.com/health
Down since Feb 1, 2026 at 09:04
Duration: 5 hours 28 minutes
Resolve

Resolved

www.mysite.com
Jan 28, 14:45 → Jan 28, 14:52 (7 min) · Resolved by alice@company.com

api.example.com/health
Jan 25, 09:04 → Jan 25, 09:14 (10 min) · Resolved by john@company.com

staging.mysite.com
Jan 20, 17:30 → Jan 20, 18:45 (1h 15min) · Resolved by john@company.com

Load more ↓

Link to diagram (made using figma): https://www.figma.com/site/CFWRT6lpKymj6Xml8Yv2gK/Internet-tech-project?node-id=0-1&t=8FYUEBdoJ2UNnWUf-1

Wireframe 5

/team

upty.app/team

Upty

Dashboard

Incidents

Team

Settings

Company Inc

3 members

JD — john@company.com — Joined Jan 15, 2026

AL — alice@company.com — Joined Jan 20, 2026

BO — bob@company.com — Joined Feb 1, 2026

Invite a teammate

Generate Invite Link

After clicking:

https://upty.app/invite/a8f3k2×9...

Copy

Expires in 7 days

Link to diagram (made using figma): https://www.figma.com/site/CFWRT6lpKymj6Xml8Yv2gK/Internet-tech-project?node-id=0-1&t=8FYUEBdoJ2UNnWUf-1

Accessibility Plan

Criteria For Success

1.4.3 Contrast (Minimum) (Level AA)
Text and status indicators will meet a minimum contrast ratio of 4.5:1. UP (#28a745 on white) and DOWN (#dc3545 on white) statuses meet contrast requirements and will not rely on colour alone. Text labels will always be present.

2.1.1 Keyboard Accessible (Level A)
All core functionality (things like navigation, monitor management, team actions) will be fully usable via keyboard without requiring a mouse.

2.4.7 Focus Visible (Level AA)
All interactive elements (buttons, links, form fields, table rows) will display a clearly visible focus indicator when navigated via keyboard. This may include a highlighted outline or colour change that contrasts with the background, helping users understand their current position on the page.

3.3.1 Error Identification (Level A)
Input errors (e.g. invalid URLs or login failures) will be clearly identified near the relevant field. Messages will explain what went wrong using simple, non-technical language and provide guidance on how to correct the error.

4.1.3 Status Messages (Level AA)
Important status updates, such as monitor state changes or incident notifications, will be presented directly within the interface (e.g. dashboard or incidents page). These updates will be available to assistive technologies without relying on pop-ups or requiring additional user interaction.

APPENDIX

Simran Singh contributed 33%
Rory Grant contributed 33%
Ying GUO contributed 33%

Each member contributed equally.